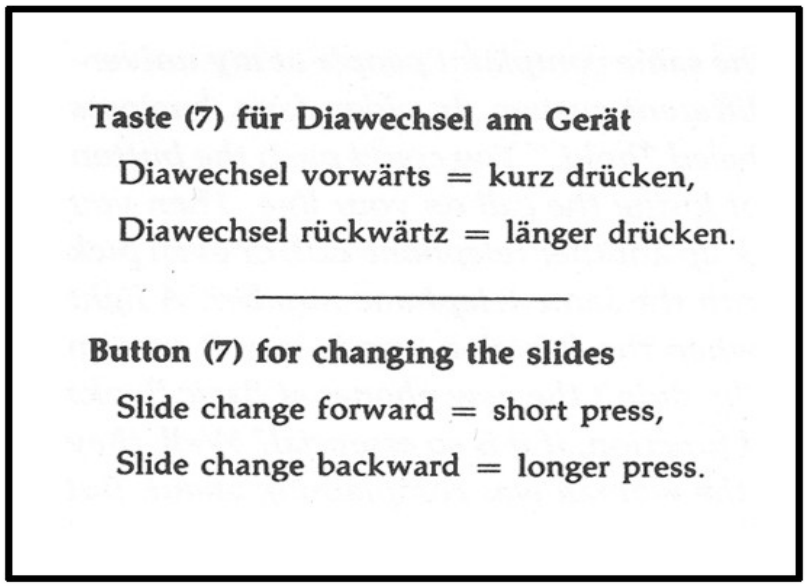
# THE DESIGN OF EVERYDAY THINGS

BY DONALD A. NORMAN

**Figure 1**

**Leitz Pravodit Slide Projector. I finally tracked down the instruction manual for that projector. A photograph of the projector has its parts numbered. The button for changing slides is number 7. The button itself has no labels. Who could discover this operation without the aid of the manual? Here is the entire text related to the button, in the original German and in my English translation.**



**Taste (7) für Diawechsel am Gerät**

Diawechsel vorwärts = kurz drücken,

Diawechsel rückwärtz = länger drücken.

---

**Button (7) for changing the slides**

Slide change forward = short press,

Slide change backward = longer press.

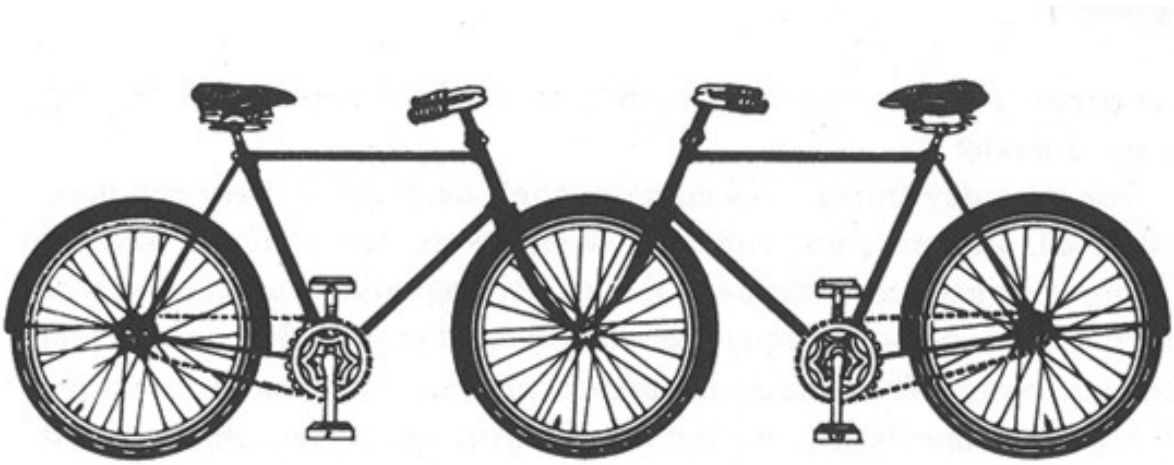
**Figure 2**

**Plate Mounted Over the Dial of the Telephones at the University of Michigan. These inadequate instructions are all that most users see. (The button labeled "TAP" at the lower right is used to transfer or pick up calls—it is pressed whenever the instruction plate says "TAP." The light on the lower left comes on whenever the telephone rings.)**



**Figure 3**

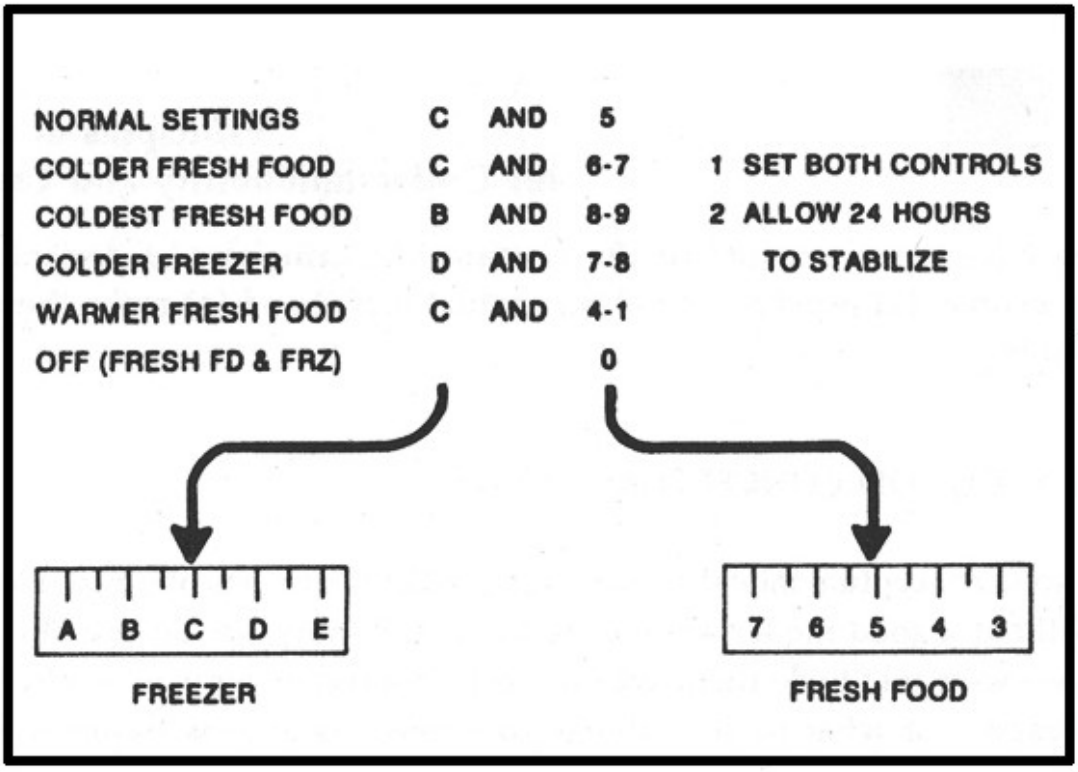
**Carelman's Tandem “Convergent Bicycle (Model for Fiancés).”**



Jacques Carelman: “Convergent Bicycle.” Copyright 1969-76-80 by Jacques Carelman and A.D.A.G.P. Paris. From Jacques Carelman, *Catalog of Unfindable Objects*, Balland, éditeur, Paris-France. Used by permission of the artist.

**Figure 4**

**My Refrigerator. Two compartments—fresh food and freezer—and two controls (in the fresh food unit). The illustration shows the controls and instructions. Your task: Suppose the freezer is too cold, the fresh food section just right. How would you adjust the controls so as to make the freezer warmer and keep the fresh food the same?**




From Norman, D. A. (1986). Cognitive engineering. In D. A. Norman & S. W. Draper (Eds.), *User centered system design: New perspectives on human-computer interaction*. Hillsdale, NJ: Erlbaum Associates.

**Figure 5**

**Two Conceptual Models for My Refrigerator. The model *A* is provided by the system image of the refrigerator as gleaned from the controls and instructions; *B* is the correct conceptual model. The problem is that it is impossible to tell in which compartment the thermostat is located and whether the two controls are in the freezer and fresh food compartment, or vice versa.**

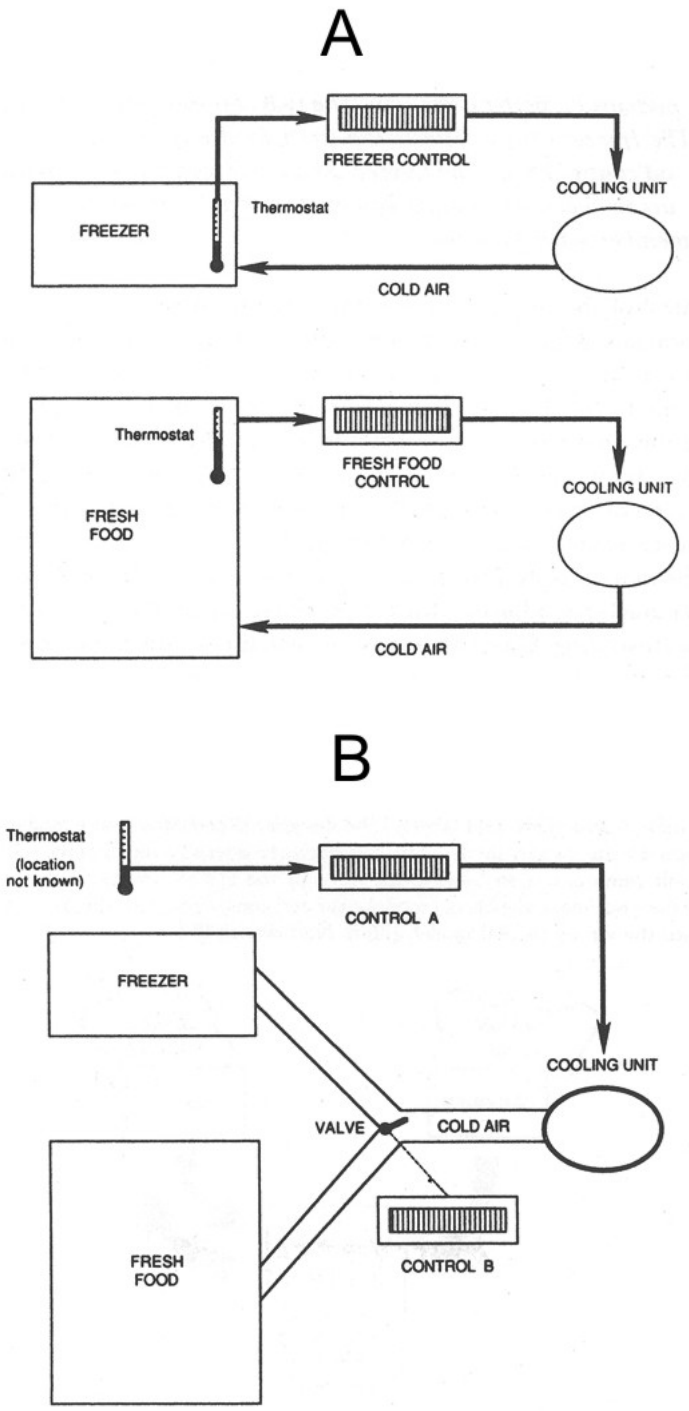
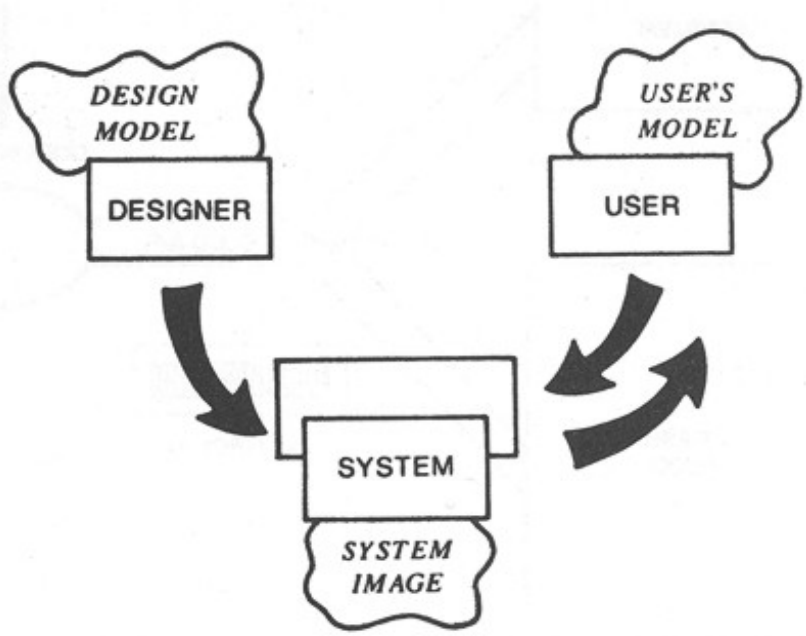


**Figure 6**

**Conceptual Models. The *design model* is the designer's conceptual model. The *user's model* is the mental model developed through interaction with the system. The *system image* results from the physical structure that has been built (including documentation, instructions, and labels). The designer expects the user's model to be identical to the design model. But the designer doesn't talk directly with the user—all communication takes place through the system image. If the system image does not make the design model clear and consistent, then the user will end up with the wrong mental model.**

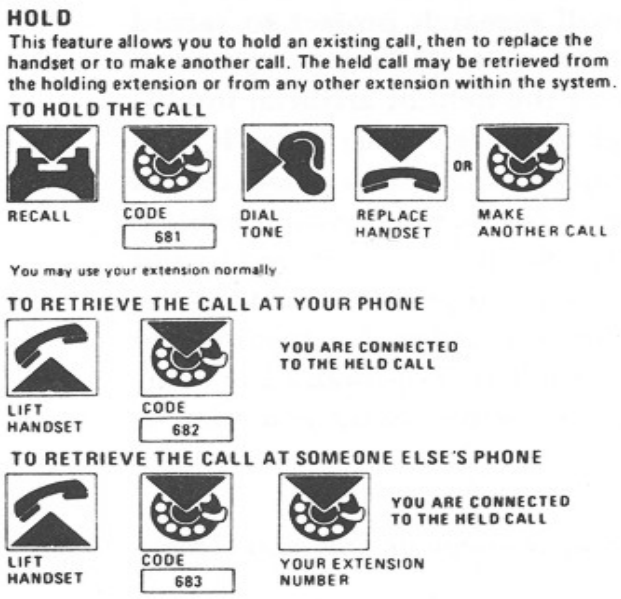



From Norman, D. A. (1986). Cognitive engineering. In D. A. Norman & S. W. Draper (Eds.), *User centered system design: New perspectives on human-computer interaction*. Hillsdale, NJ: Erlbaum Associates.

**Figure 7**

**Two Ways to Use Hold on Modern Telephones. Illustration *A* is the instruction manual page for British Telecom. The procedure seems especially complicated, with three 3-digit codes to be learned: 681, 682, and 683. Illustration *B* shows the equivalent instructions for the Ericsson Single Line Analog Telephone installed at the University of California, San Diego. I find this set of instructions easier to understand, but one must still dial an arbitrary digit: 8 in this case.**

A




B

**CALL HOLD/CALL PARK**

With party on line

- Press **R** key
- Listen for recall dial tone (three beeps and dial tone)
- Hang up handset

TO RETRIEVE FROM SAME PHONE

- Lift handset; you are connected to the call

TO RETRIEVE FROM ANOTHER PHONE

- Lift handset
- Dial extension where call was parked; listen for busy tone
- Dial **8**; you are connected to the call

**NOTE:** Call will remain parked for 3 minutes before re-ringing

**Figure 8**

**Seat Adjustment Control from a Mercedes-Benz Automobile. This is an excellent example of natural mapping. The control is in the shape of the seat itself: the mapping is straightforward. To move the front edge of the seat higher, lift up on the front part of the button. To make the seat back recline, move the button back. Mercedes-Benz automobiles are obviously not everyday things for most people, but the principle doesn't require great expense or wealth. The same principle could be applied to much more common objects.**

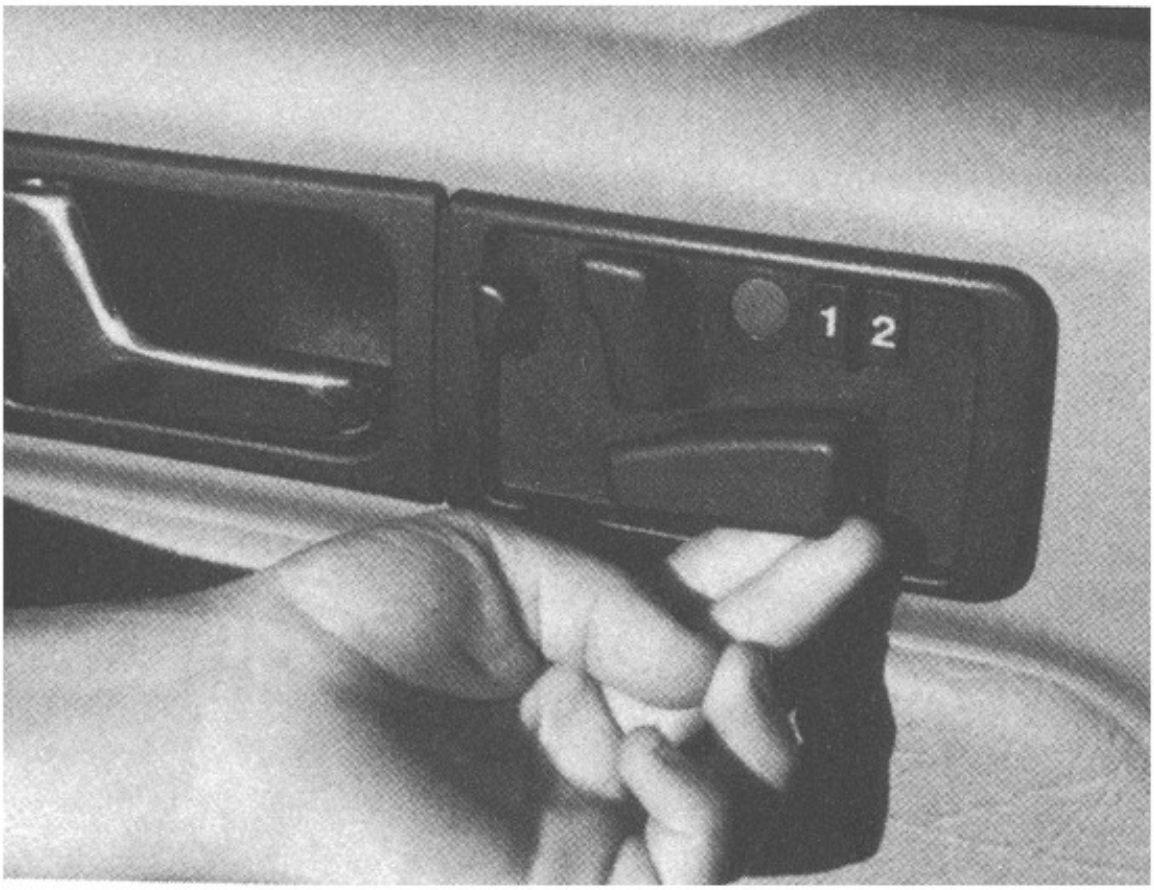
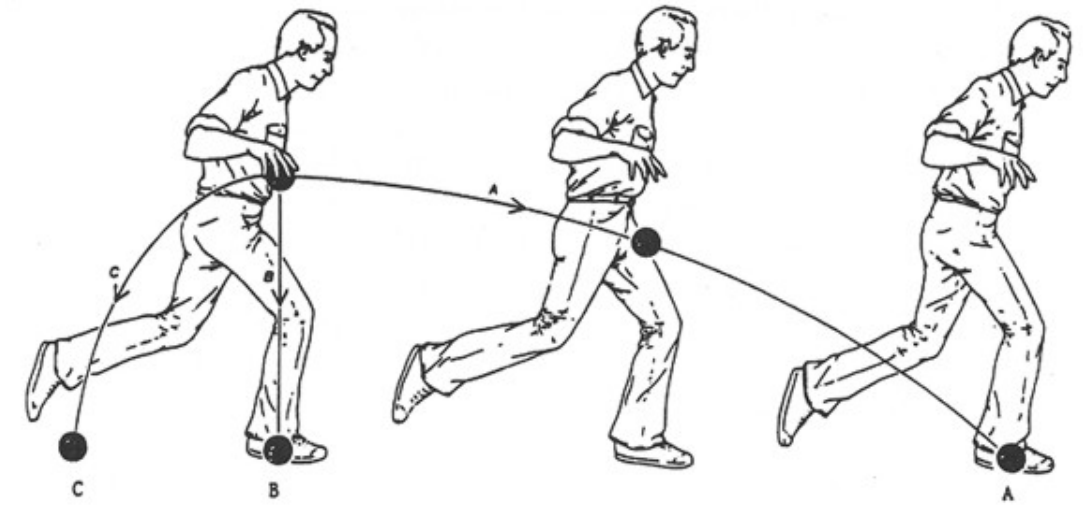


**Figure 9**

**A Running Man Drops a Ball. Which path does the ball take as it falls to the ground, path A, B, or C? When this question was asked of sixth-grade students in Boston schools, only 3 percent answered A, the right answer; the others were evenly divided between B and C. Even high school students did not do well: of forty-one students who had just studied Newtonian mechanics for a month and a half, only 20 percent got the right answer; the others were almost equally divided between B and C.**




The study was performed by White & Horwitz, 1987. The figure is reprinted from *Intuitive Physics* by McCloskey. Copyright 1983 by *Scientific American*, Inc. All rights reserved.

**Figure 10**

**The Action Cycle. Human action has two aspects, execution and evaluation. Execution involves doing something. Evaluation is the comparison of what happened in the world with what we wanted to happen (our goal).**

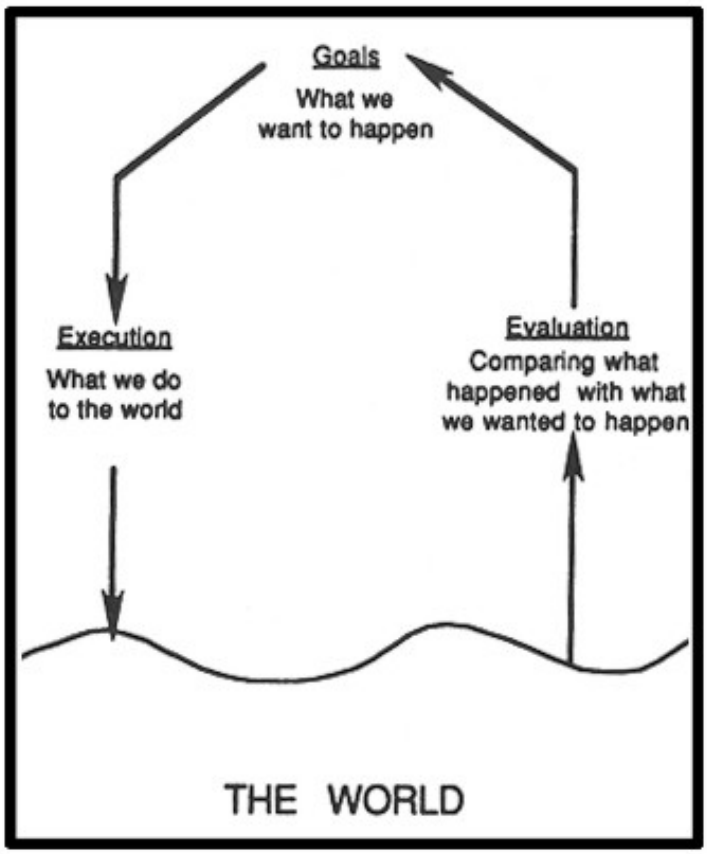


**Figure 11**

**Stages of Execution. Start at the top with the *goal*, the state that is to be achieved. The goal is translated into an *intention* to do some action. The intention must be translated into a set of internal commands, an *action sequence* that can be performed to satisfy the intention. The action sequence is still a mental event: nothing happens until it is *executed*, performed upon the world.**

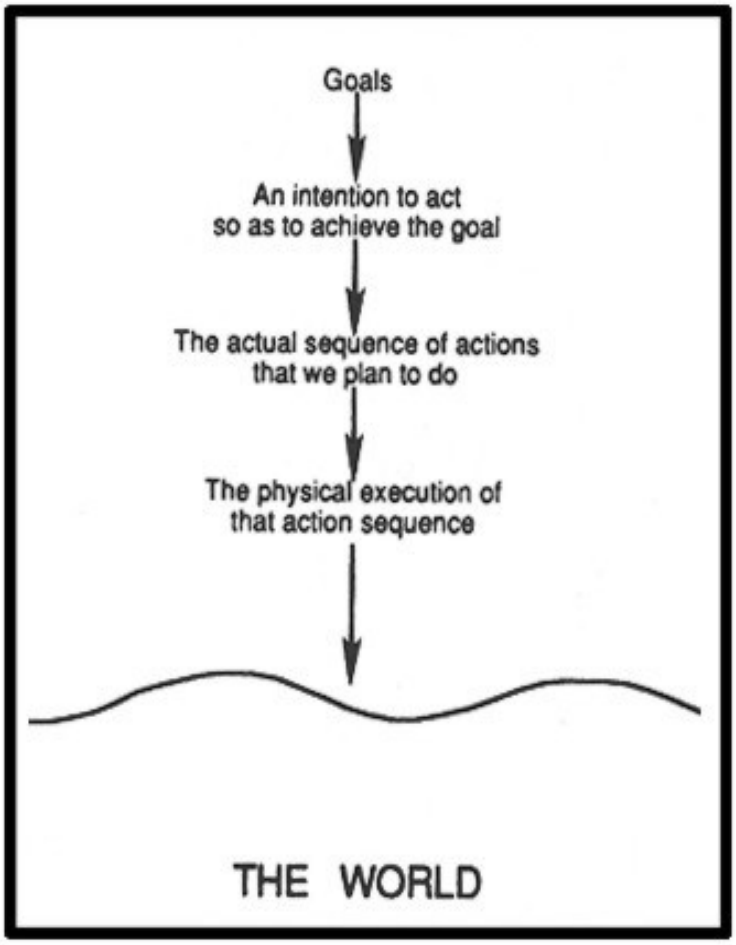


**Figure 12**

**Stages of Evaluation. Evaluation starts with our *perception* of the world. This perception must then be *interpreted* according to our expectations and then compared (*evaluated*) with respect to both our intentions (from figure 11) and our goals.**

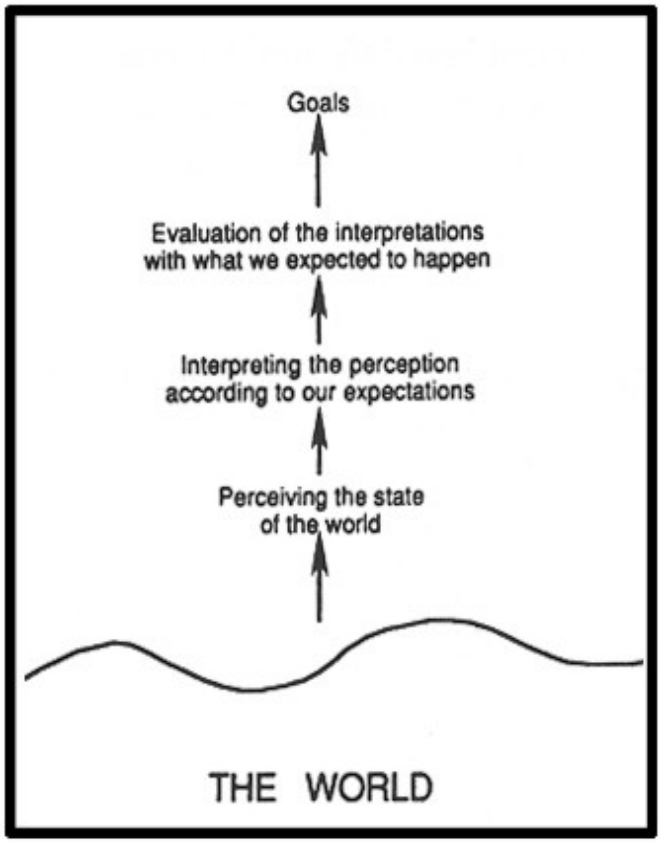


**Figure 13**

**Seven Stages of Action. The stages of execution from figure 11 (intentions, action sequence, and execution) are coupled with the stages of evaluation from figure 12 (perception, interpretation, and evaluation), with goals common to both stages.**

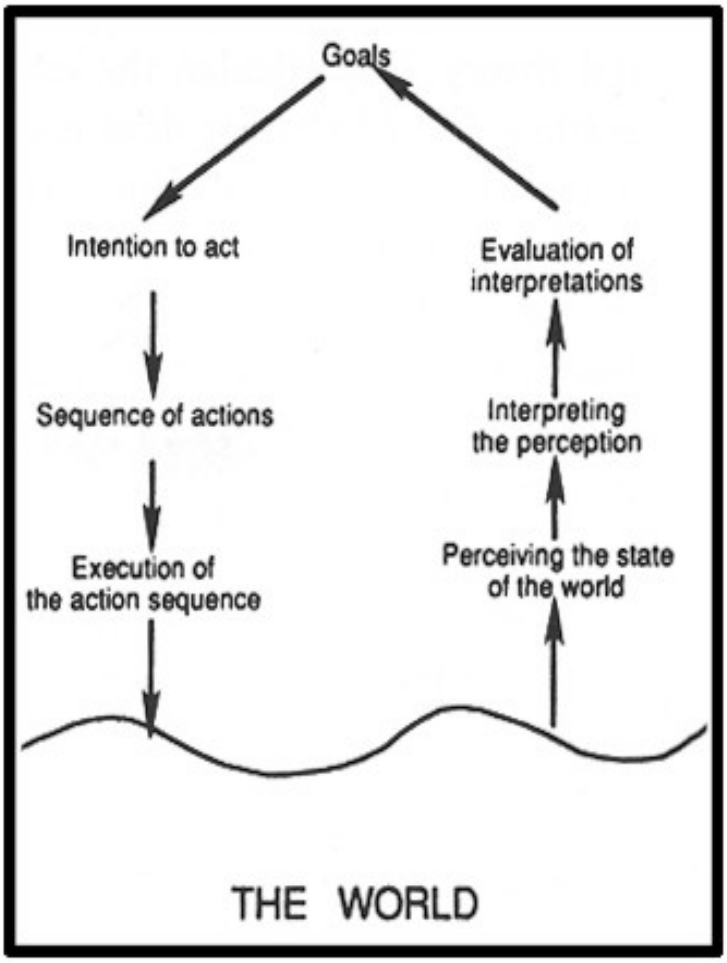
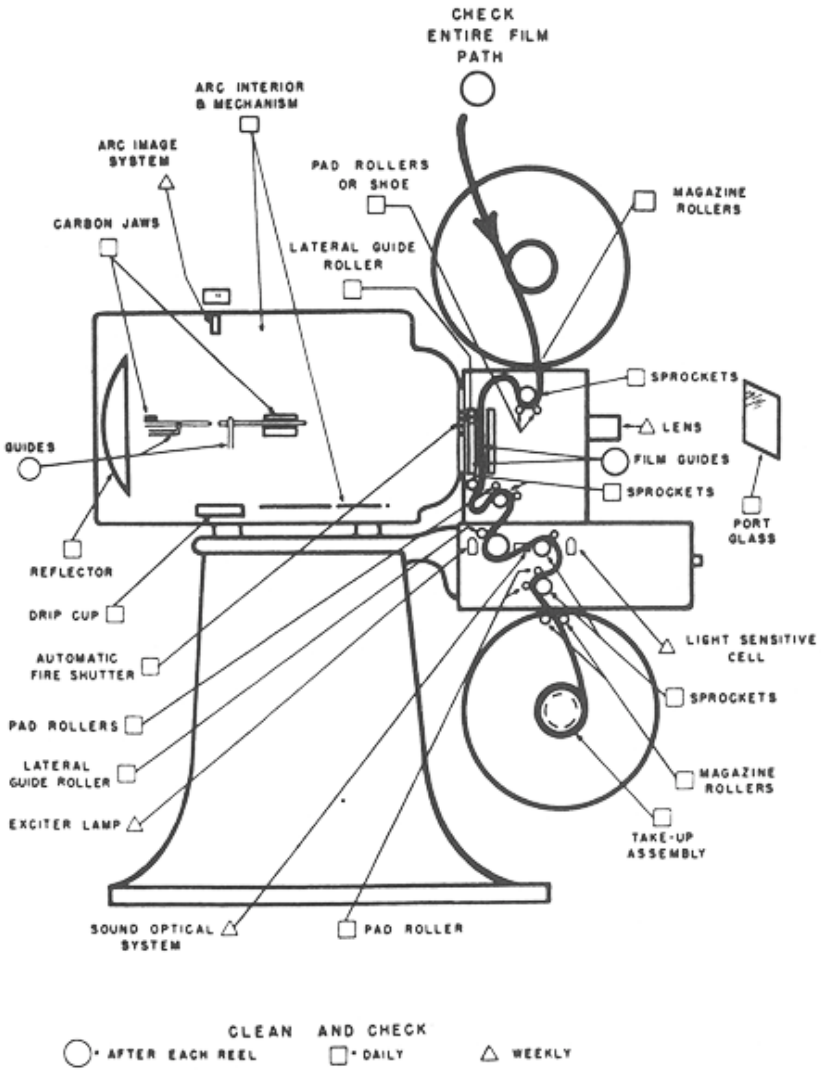


**Figure 14**

**Threading the Movie Projector. The dark line at the right shows the path of the film. This picture doesn't tell the whole story, for the several loops of film have to be threaded just right, neither too loose nor too taut.**




From *Projectionist's manual*, Department of the Army and the Air Force, May 1966.

**Figure 15**

**Using the Seven Stages to Ask Design Questions**

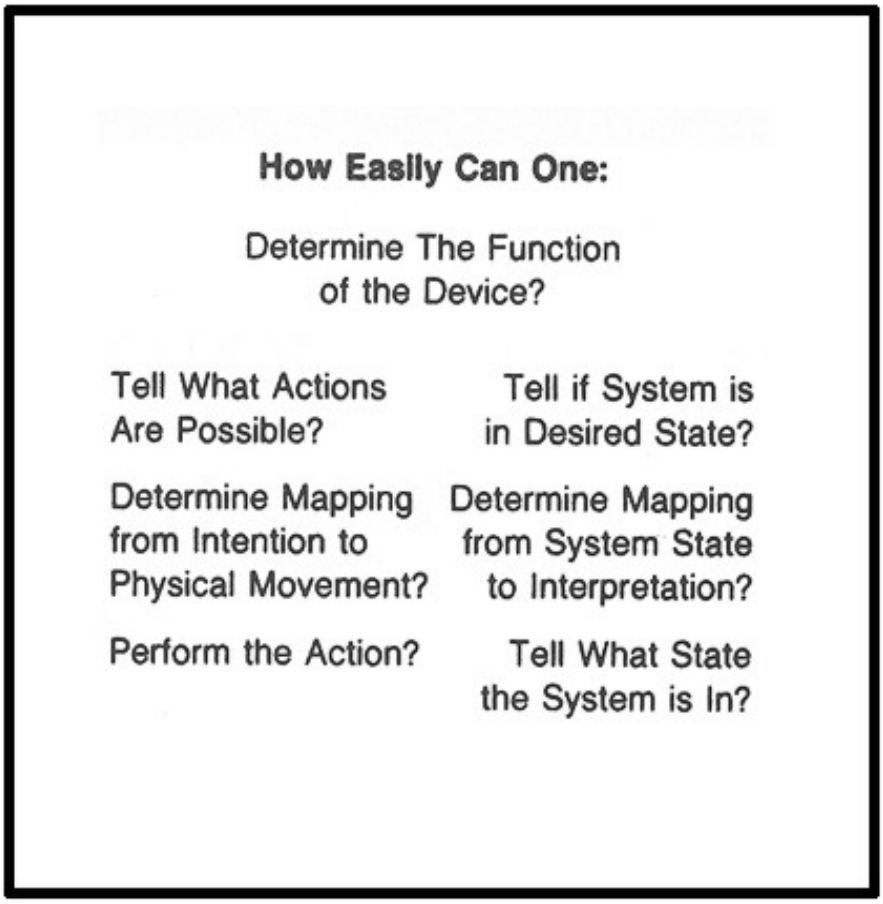
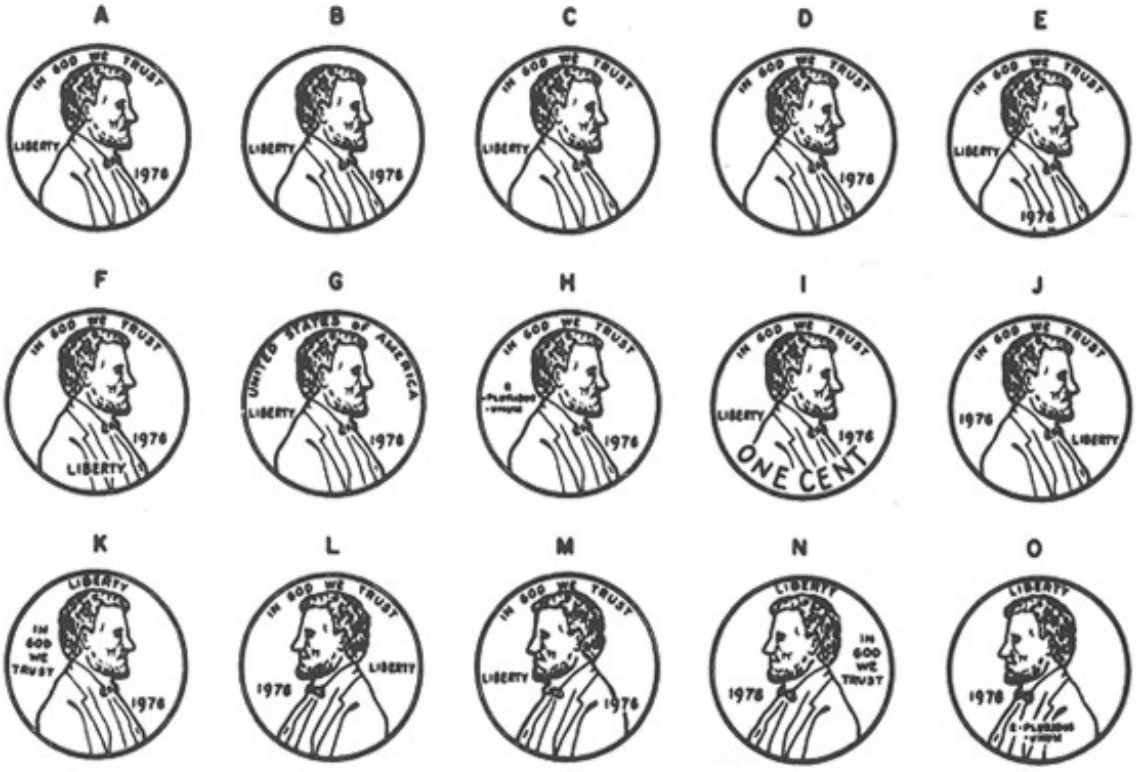


**Figure 16**

**Which is the U.S. One Cent Coin—The Penny? Fewer than half of the American college students who were given this set of drawings and asked to select the correct one could do so. Pretty bad performance, except that the students, of course, have no difficulty using the money: in normal life, we have to distinguish between the penny and other U.S. coins, not between several versions of one denomination.**




From Nickerson & Adams, *Cognitive Psychology, 11,* copyright 1979.
Reprinted by permission of Academic Press.

**Figure 17**

**Arbitrary Arrangement of Stove Controls. Couple the usual rectangular arrangement of burners with this arbitrary row of controls, and there is trouble: which control goes with which burner? You don't know unless the controls are labeled. The memory load for this arrangement is high: there are twenty-four possible arrangements, and you have to remember which of the twenty-four this one is. Fortunately, the controls are seldom arranged quite this arbitrarily.**

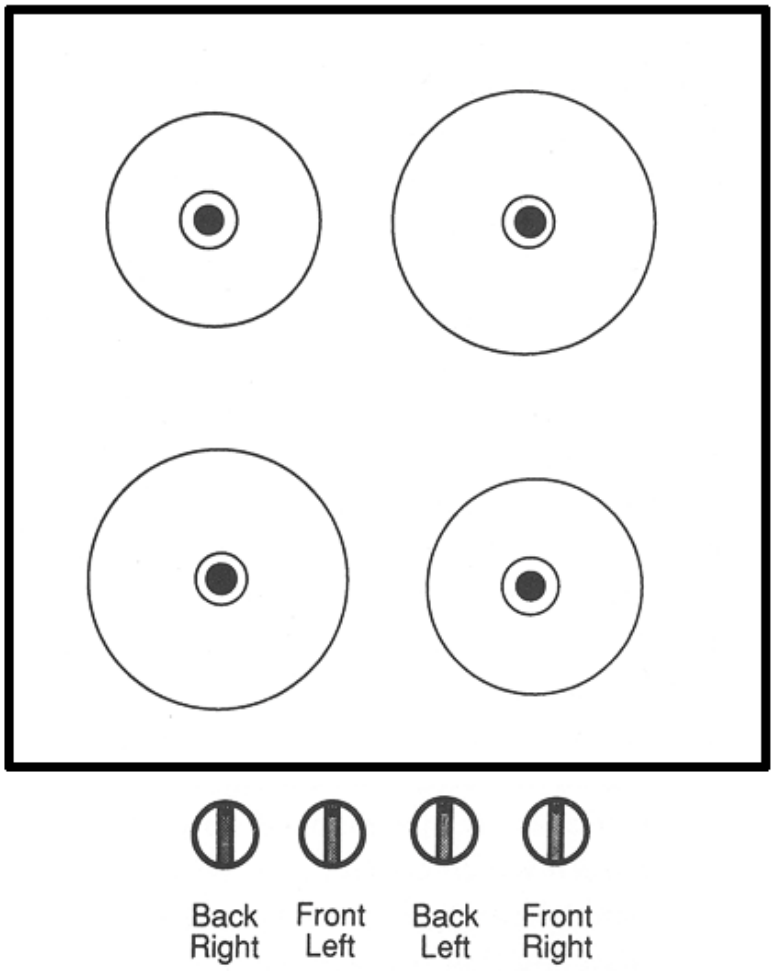


**Figure 18**

**Paired Stove Controls. This is the type of partial mapping of controls to burners in common use today. The two controls on the left work the left burners, and the two controls on the right work the right burners. Now there are only four possible arrangements (two for each side). Even so, confusion is possible (and, I can assure you, it occurs often).**

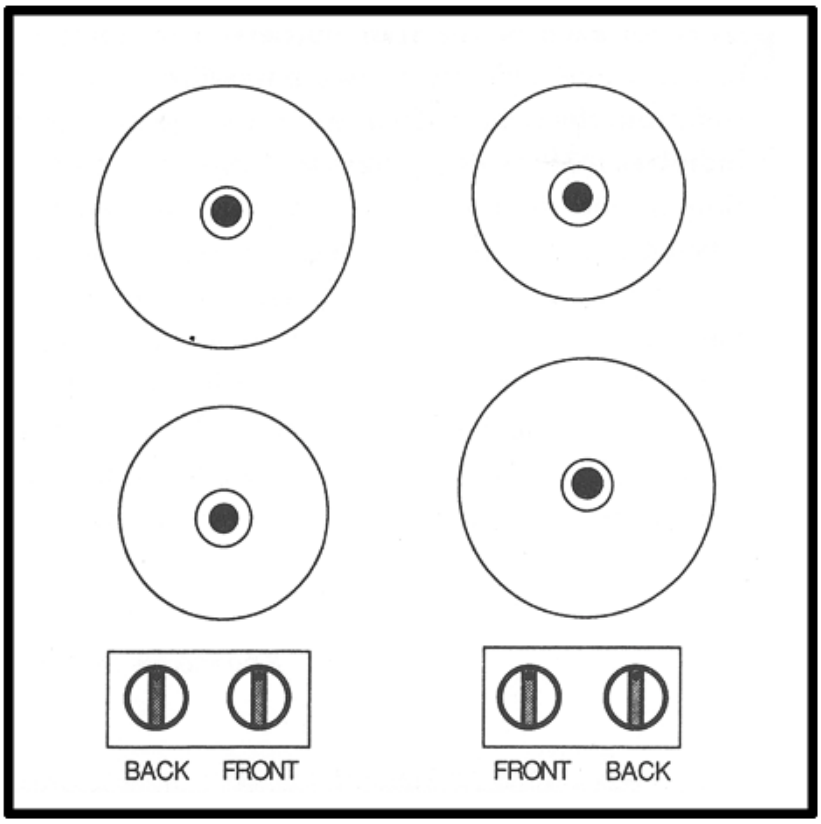


**Figure 19**

**Full Natural Mapping of Controls and Burners. Two of the Possible Ways. There is no ambiguity, no need for learning or remembering, no need for labels. Why can't all stoves be like these?**

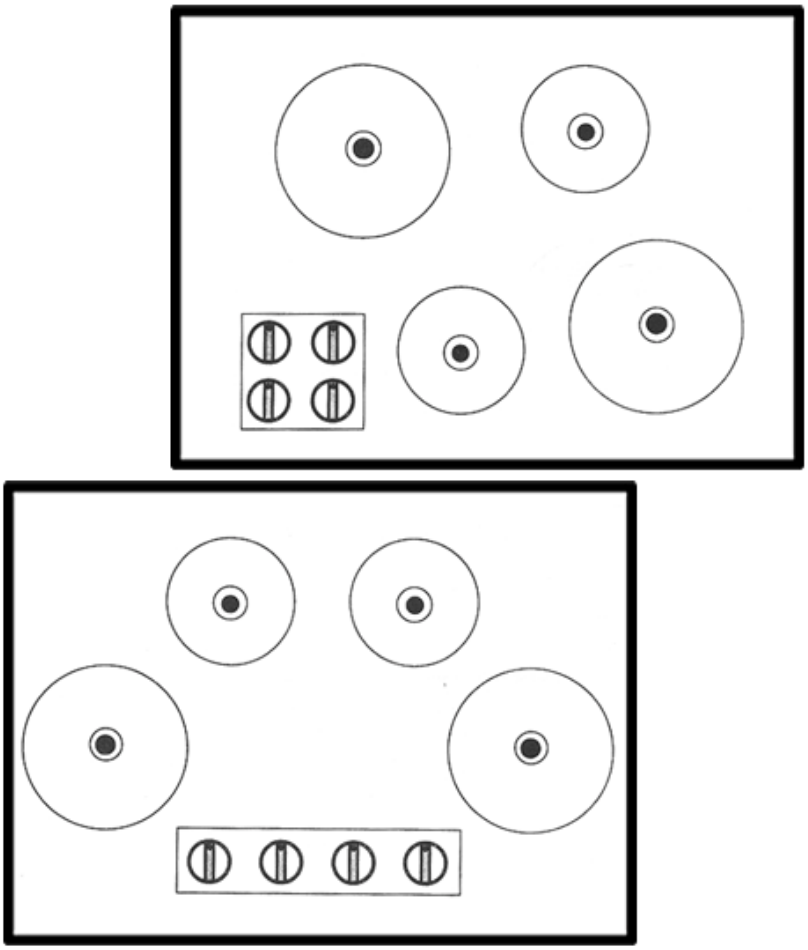


**Figure 20**

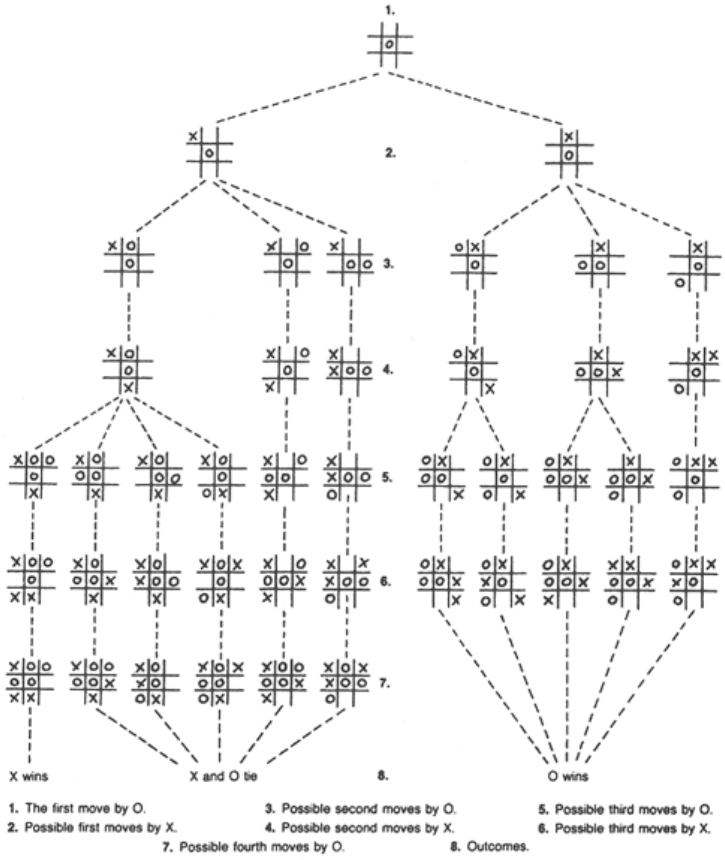
**The Tradeoff between Knowledge in the World and in the Head**

**Tradeoffs**

| PROPERTY | KNOWLEDGE IN THE WORLD | KNOWLEDGE IN THE HEAD |
|---|---|---|
| *Retrievability* | Retrievable whenever visible or audible. | Not readily retrievable. Requires memory search or reminding. |
| *Learning* | Learning not required. Interpretation substitutes for learning. How easy it is to interpret information in the world depends upon how well it exploits natural mappings and constraints. | Requires learning, which can be considerable. Learning is made easier if there is meaning of structure to the material (or if there is a good mental model). |
| *Efficiency of use* | Tends to be slowed up by the need to find and interpret the external information. | Can be very efficient. |
| *Ease of use at first encounter* | High. | Low. |
| *Aesthetics* | Can be unaesthetic and inelegant, especially if there is a need to maintain a lot of information. This can lead to clutter. In the end, aesthetic appeal depends upon the skill of the designer. | Nothing need be visible, which gives more freedom to the designer, which in turn can lead to better aesthetics. |

**Figure 21**

**Wide and Deep Decision Tree. The game of tic-tac-toe (naughts and crosses). The tree starts at the top, with the initial state, then deepens as each successive layer considers all the alternative moves by each player. Although this diagram looks a bit complex, it is a pretty simple structure as these things go. First of all, this picture is much simplified. Only one possible first move by O is shown, and the symmetry of the board is used to reduce the number of alternatives being considered. (Only two first moves by X need be considered: the eight possibilities are really equivalent to the two shown because of the symmetry.) In the full game, there are nine possible first moves for O, eight possible replies by X, seven second moves by O, and so on, up to the third move by O, which is the first possible time for the game to be won; there are 15,120 possible sequences up to that point. Even this simple game leads to such a wide and deep decision tree that it is not possible to work out all the possibilities in the head. Expert players take advantage of simple strategies and memorized move sequences.**




From *Human Information Processing,* Second Edition, by Peter H. Lindsay and Donald A. Norman, copyright 1977 by Harcourt Brace Jovanovich, Inc. Reprinted by permission of the publisher.

**Figure 22**

**The Nintendo Children's Toy. This home video game set is intended for use by children. However, it has a complex safety instruction, one almost guaranteed to be ignored. To use the system, one inserts a "game pak" cartridge into the "chamber." The power switch should be off when inserting or removing the cartridge. In the absence of any forcing function, the instruction is almost universally disregarded (if anyone even knows about it). If order is important, there should be a forcing function. If order does not matter, the instruction should be dropped.**

## OPERATING YOUR NES

●TO START PLAY

1. Turn your television on to Channel 3.

   Note: If Channel 3 is broadcasting in your area and interfering with the game, set the switch on the back of the Control Deck to Channel 4.

2. If your TV has an automatic fine tuning control (AFC), turn it off. (Use the manual fine tune dial to adjust the picture after inserting the game pak as described below.)

   Note: If you have a color TV that turns black and white when the AFC is turned off, leave the AFC on.

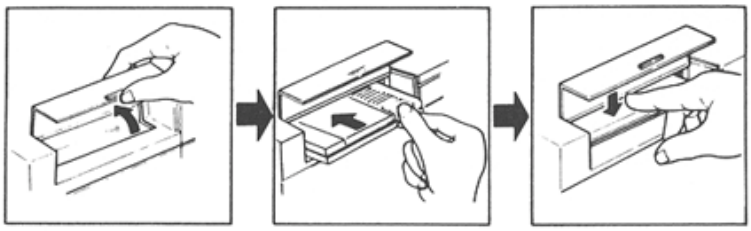
3. Make sure that the power switch on the Control Deck is off.

CAUTION !! ALWAYS MAKE SURE THAT THE POWER SWITCH ON THE CONTROL DECK IS OFF BEFORE INSERTING OR REMOVING A GAME PAK !!

4. Open the Chamber Lid on the Control Deck.

Insert a Game Pak into the Chamber (Label Facing up) and Push it all the way in.

Press Down on the Game Pak until it locks into place and close the Chamber Lid.

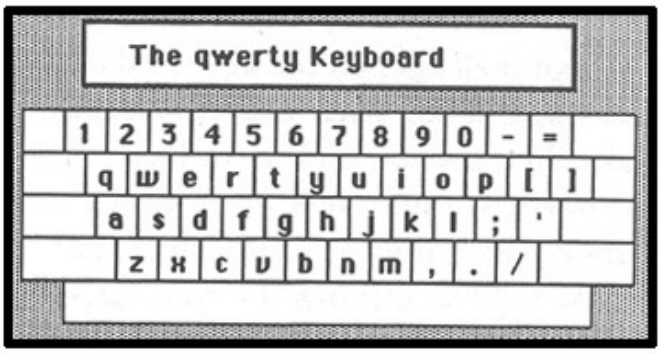


From the Nintendo instruction manual. Nintendo and Nintendo Entertainment System are trademarks of Nintendo of America Inc. Copyright 1986 Nintendo.

**Figure 23**

**Typewriter Keyboards.**

**The standard American layout of keys—the Sholes or qwerty keyboard.**




**The American Simplified Keyboard (often called ASK), a simplified version of the original Dvorak keyboard; on the original, the numerals and punctuation keys are arranged differently.**

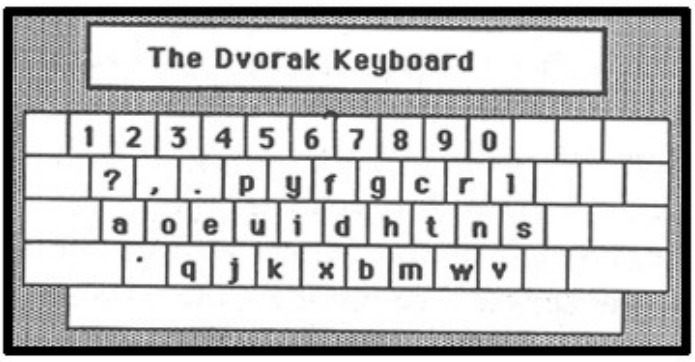
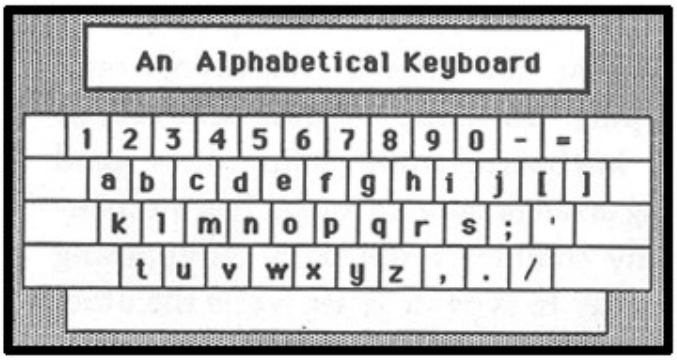


**Figure 23, continued**

**Most alphabetically organized keyboards arrange the alphabet along horizontal rows, as shown.**




**This alphabetical arrangement is superior, however: with its diagonal arrangement, letters increase systematically up the alphabet from left to right without major breaks.**

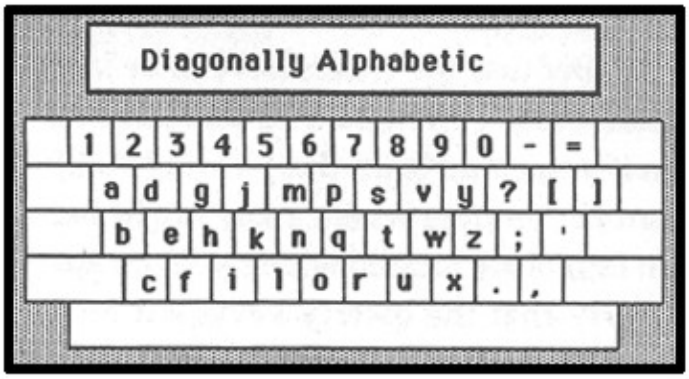


**Figure 23, continued**

**This keyboard has randomly arranged letters.**




**Beginners succeed about the same on all these keyboards: alphabetical works barely better than random. For experts, ASK is best, followed by qwerty: alphabetical keyboards are quite inferior. Moral: Don’t bother with alphabetical keyboards.**

**Figure 24**

**Vertical faucets. The world standard is that hot is on the left, cold on the right. What do you do here? Why would anyone dream up this scheme?**

